



NeuroSpace

Complete evaluation export

Job ID	cb96c72d-54d3-4c68-a6a3-1613d26a6de4
Status	completed
Result timestamp	2026-05-13T19:57:41.975658
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Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	30
manual_overrides.rnn_batch_size	64
manual_overrides.qrnn_epochs	15
manual_overrides.qrnn_batch_size	16
num_classes	6
class_names	1, 2, 3, 4, 5, 6
qrnn_backend	qiskit
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1

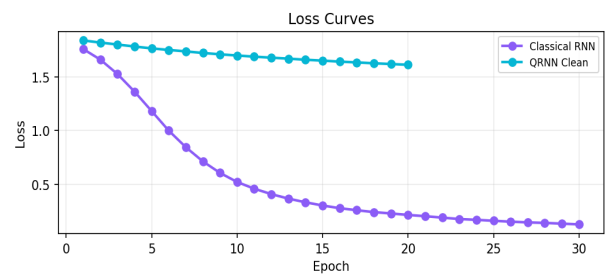
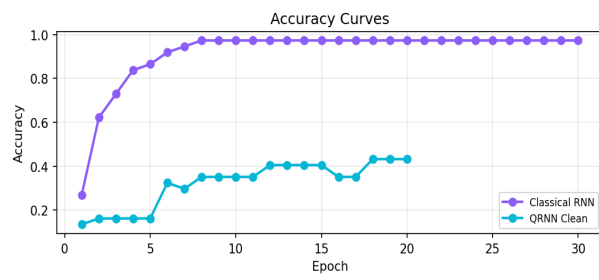
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	91.89%	47.30%	44.59%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	91.89%	93.55%	91.89%	91.78%	0.1314
QRNN Clean	47.30%	52.20%	47.30%	45.72%	1.6103

Learning Curves



Classical RNN

Metric	Value
Accuracy	91.89%
Precision	93.55%
Recall	91.89%
F1 Score	91.78%
Loss	0.1314
Train Loss	0.1336
Val Loss	0.1314
Train Accuracy	98.04%
Val Accuracy	97.30%
Train Size	255
Val Size	37
Test Size	74

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1				
Class 0	22	0	0	0	1	0
Class 1	0	8	0	2	1	1
Class 2	0	0	14	1	0	0
Class 3	0	0	0	10	0	0
Class 4	0	0	0	0	10	0
Class 5	0	0	0	0	0	4

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
1	100.00%	95.65%	97.78%	23.0000
macro avg	90.04%	92.61%	90.18%	74.0000
weighted avg	93.55%	91.89%	91.78%	74.0000
Accuracy			91.89%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8605	1.7558	22.35%	27.03%
2	1.5483	1.6579	42.75%	62.16%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
3	1.2854	1.5246	59.61%	72.97%
4	1.1306	1.3613	73.33%	83.78%
5	0.9797	1.1813	78.82%	86.49%
6	0.8686	1.0002	84.31%	91.89%
7	0.7719	0.8466	87.06%	94.59%
8	0.6668	0.7147	89.80%	97.30%
9	0.6322	0.6083	90.59%	97.30%
10	0.5620	0.5242	91.37%	97.30%
11	0.4921	0.4620	92.55%	97.30%
12	0.4623	0.4121	93.73%	97.30%
13	0.4168	0.3700	92.94%	97.30%
14	0.3396	0.3362	95.29%	97.30%
15	0.3426	0.3062	94.12%	97.30%
16	0.3125	0.2814	95.29%	97.30%
17	0.3058	0.2627	96.08%	97.30%
18	0.2679	0.2451	96.08%	97.30%
19	0.2330	0.2327	97.25%	97.30%
20	0.2532	0.2189	96.47%	97.30%
21	0.2327	0.2075	94.90%	97.30%
22	0.1977	0.1939	97.65%	97.30%
23	0.2112	0.1813	95.69%	97.30%
24	0.1799	0.1733	97.25%	97.30%
25	0.1653	0.1649	97.65%	97.30%
26	0.1494	0.1566	98.04%	97.30%
27	0.1408	0.1499	98.04%	97.30%
28	0.1424	0.1451	98.04%	97.30%
29	0.1426	0.1384	98.04%	97.30%
30	0.1336	0.1314	98.04%	97.30%

QRNN Clean

Metric	Value
Accuracy	47.30%
Precision	52.20%
Recall	47.30%
F1 Score	45.72%
Loss	1.6103
Train Loss	1.5954
Val Loss	1.6103
Train Accuracy	44.71%
Val Accuracy	43.24%
Train Size	255
Val Size	37
Test Size	74

QRNN Clean Confusion Matrix

Actual / Predicted	Class 0	Class 1				
Class 0	14	5	1	2	1	0
Class 1	0	6	0	5	1	0
Class 2	0	8	1	2	4	0
Class 3	0	3	0	6	1	0
Class 4	0	0	2	0	8	0
Class 5	0	3	0	0	1	0

QRNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
1	100.00%	60.87%	75.68%	23.0000
macro avg	39.83%	42.92%	38.03%	74.0000
weighted avg	52.20%	47.30%	45.72%	74.0000
Accuracy			47.30%	

QRNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8534	1.8357	18.43%	13.51%
2	1.8238	1.8162	18.82%	16.22%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
3	1.8046	1.7980	18.82%	16.22%
4	1.7855	1.7797	19.22%	16.22%
5	1.7668	1.7622	21.96%	16.22%
6	1.7480	1.7479	27.45%	32.43%
7	1.7295	1.7343	37.65%	29.73%
8	1.7192	1.7198	36.08%	35.14%
9	1.7028	1.7074	35.29%	35.14%
10	1.6922	1.6963	35.29%	35.14%
11	1.6838	1.6857	37.25%	35.14%
12	1.6677	1.6762	38.04%	40.54%
13	1.6596	1.6671	37.65%	40.54%
14	1.6489	1.6572	38.43%	40.54%
15	1.6361	1.6489	39.22%	40.54%
16	1.6297	1.6396	38.43%	35.14%
17	1.6226	1.6324	40.78%	35.14%
18	1.6134	1.6243	45.88%	43.24%
19	1.6067	1.6173	45.49%	43.24%
20	1.5954	1.6103	44.71%	43.24%